

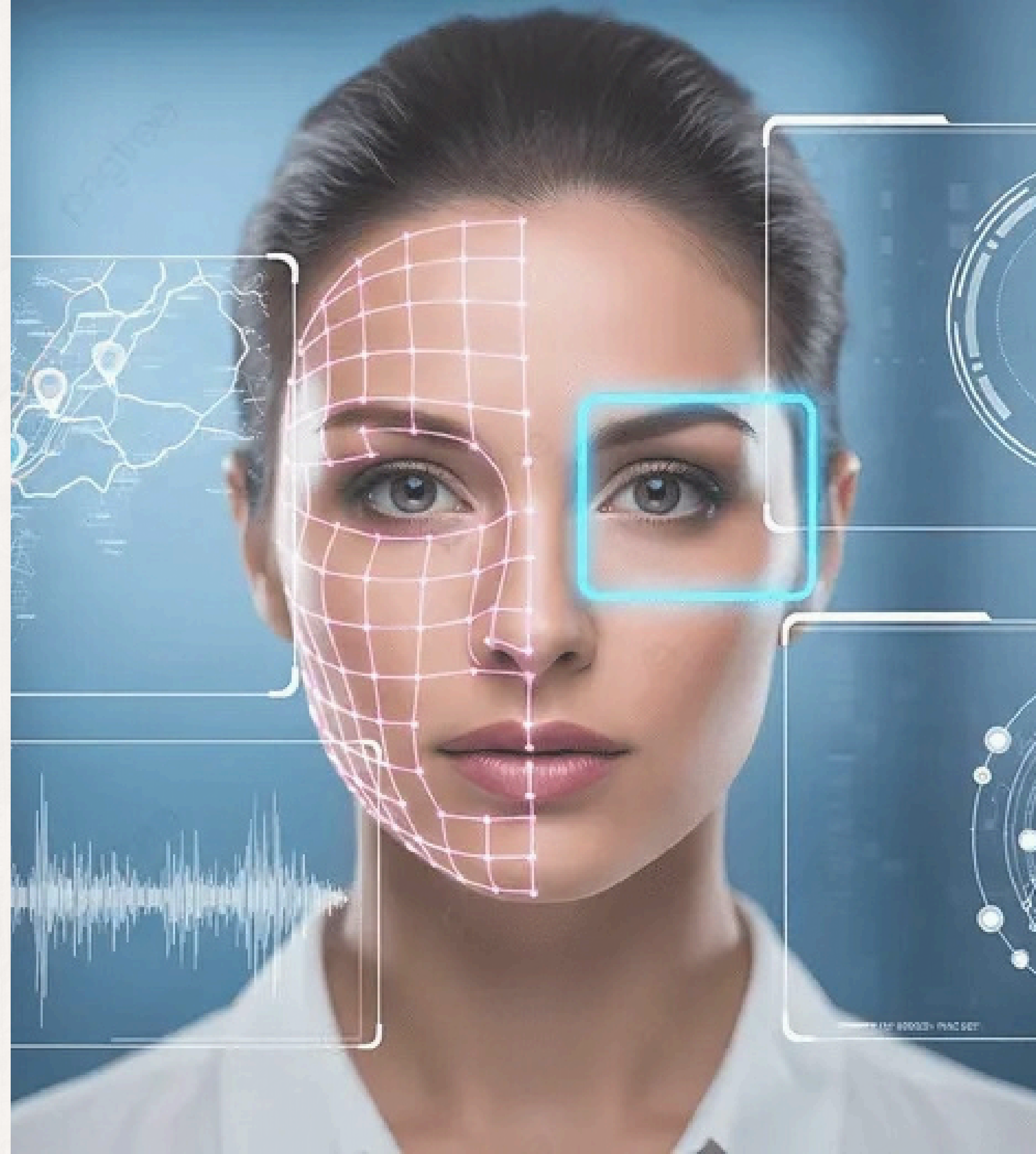
THIRD I

A smart model that helps you notice what you may not notice.

Presented By HEXORA Team

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PROBLEM STATEMENT

People often overlook gradual and subtle changes in their appearance or movement because they see themselves every day, creating a gap between the beginning of a visible change and awareness that it may require professional medical attention.

SOLUTION:

An AI-assisted personal change-monitoring system that analyzes existing photos/videos, creates a personal baseline, and detects persistent visual or movement changes.

GOAL:

Help people notice meaningful changes earlier and encourage appropriate professional consultation — without diagnosing diseases.

KEY FEATURES & MODULES

1. Personal Baseline
2. Colour & Shape Change Detection
3. Face/Neck Geometry Analysis
4. Motion/Swallow Analysis
5. Previous vs Current Comparison
6. Persistent Change Alert
7. Generating Summary
8. Privacy-focused processing

1. Media Input → Photos/Videos
2. Quality Check → Lighting, angle, blur
3. Region Detection → Eye, skin, face, neck
4. Feature Extraction → Colour, shape, geometry, motion
5. Change Engine → Baseline comparison
6. Alert Module → Explain change + suggest consultation

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Strengths

- Personal baseline-based detection.
- Non-invasive and smartphone-based.
- Minimal disease-specific training
- Explainable change detection.
- Supports early health awareness.

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Weaknesses

- Sensitive to lighting and camera quality.
- Cannot diagnose diseases.
- Possible false alerts. Requires
- comparable media. Needs clinical validation.

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Opportunities

- Preventive healthcare
- awareness. Telehealth integration.
- Family/caregiver monitoring.
- Scalable through smartphones.
- Expandable to more health features.

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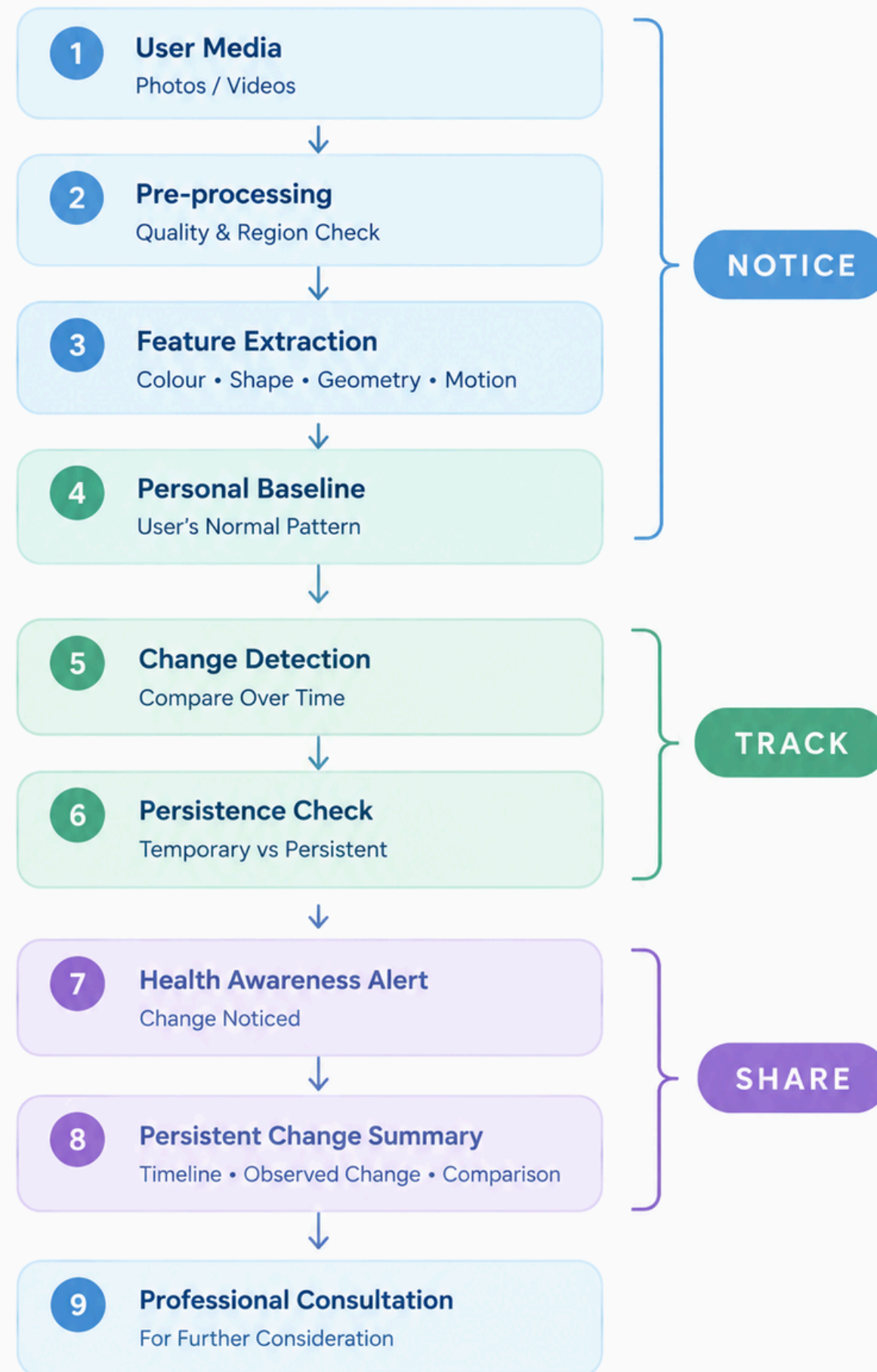
Threats

- Privacy and data-security risks.
- Misinterpretation as diagnosis.
- Over-reliance on AI.
- Regulatory requirements.
- Variation across users and devices.

Workflow

Third Eye

Helping You Notice What You May Not Notice



RISK MITIGATION

- Lighting changes → normalization
- Poor images → quality validation
- False alerts → persistence + confidence threshold
- Privacy → local processing where possible
- Misdiagnosis → clearly state “Change detection, not diagnosis”
- Temporary changes → repeated observations



Third Eye

Helping You Notice What You May Not Notice

Your everyday photos.
Our smart analysis.
A healthier tomorrow.



How It Works

1

Existing Photos / Videos

From your gallery or camera.



No extra effort.
Just your daily media.

2

Feature Analysis

Detects and measures key visual features.

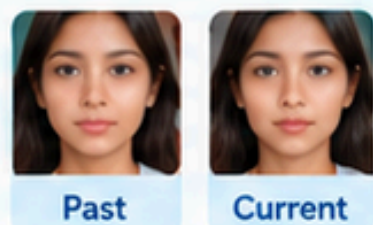


- Colour
- Shape / Geometry
- Texture
- Motion

3

Personal Baseline

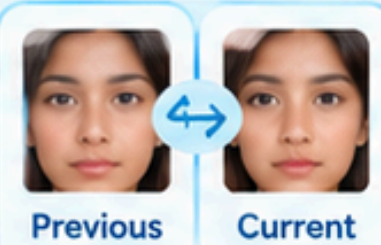
Learns what is normal for YOU.



4

Compare Over Time

Looks for differences between previous and current images.



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Change Detected

If a change is persistent, it triggers an alert.



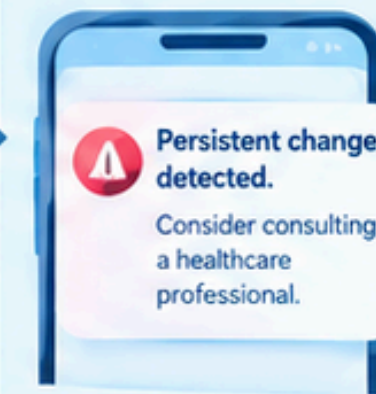
What changed?



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Health Awareness

Shows a simple alert and suggests professional evaluation.



You decide what to do next.

Examples of Changes We Can Notify



Yellowish eye/skin
→ Jaundice-related concern



Eye redness/swelling
→ Eye-related concern



Neck contour/swelling
→ Thyroid/neck concern



Skin colour/texture change
→ Skin-related concern



Swallowing movement change
→ Swallowing/neck concern



Detects changes, not diseases.
Clinical evaluation is required to determine the cause.



The Bigger Impact on Healthcare

From unnoticed change to timely care.



Change noticed
(in your photos/videos)



AI analysis
(compares with your personal baseline)



Alert
(what changed?)



Professional consultation
(clinical evaluation)



Earlier detection
(better outcomes)

— SMALL CHANGES CAN MATTER —

DATA ARCHITECTURE & API FEASIBILITY

Data Input & Collection

Source: User's existing photos/videos through Gallery/Camera APIs. Images remain on-device where possible; only necessary data is processed.

Pre-processing & Feature Extraction

OpenCV / MediaPipe: Face, eye, skin and landmark detection. Extracts colour, shape, geometry, texture and motion features. No large disease-specific image dataset required for the MVP.

Baseline & Change Analysis

Stores extracted measurements using SQLite/Firebase. Creates a personal baseline and compares new observations. Python + FastAPI can connect the computer-vision processing with the application.

Alert & User Interface

Notification API generates alerts when a persistent change is detected. Dashboard displays Previous → Current → What Changed? API/cloud services are optional; core analysis can run locally for better privacy.

“The model learns change, not disease — reducing dependence on large disease-specific datasets.”

IMPACT, SOCIAL BENEFITS & SCALABILITY

Reduction

The gap between a physical change occurring and the person noticing it.

Inclusive

Smartphone-based and non-invasive, supporting users across different age groups and backgrounds.

24/7 Monitoring

Uses existing photos/videos for continuous, passive health-change awareness.

MULTI-STAKEHOLDER BENEFITS

- Individuals – Notice & track changes
- Families – Continuous caregiver support
- Doctors – Visual history for consultation
- Healthcare Providers – Better follow-up monitoring
- Telehealth – Support remote consultation
- Project Holder – Partnerships, API & scalable health-tech platform
- Public Health – Preventive-health awareness at scale

NATIONAL SCALE & GOVT ALIGNMENT

National Scale: Smartphone-based design enables expansion from individuals → families → communities → nationwide use.

Digital Health: Can potentially integrate with India's digital-health ecosystem and telehealth services.

Government Alignment: Supports broader goals around accessible, preventive and technology-enabled healthcare.

Future Scope: Can be adapted for public-health awareness programs after clinical validation, privacy assessment and regulatory approval.



Third Eye vs. Existing Health Tech Models

Mapping our zero-effort passive tracking architecture against current medical and consumer vision benchmarks.

Clinical Screeners

Examples: BiliCam, Picterus

Relies on single-point clinical captures and physical calibration cards for neonates.

Limitation: Active, high friction

Skin & Mole Logs

Examples: Miiskin, SkinVision

Requires manual user session scheduling to track localized lesions or dermatological changes.

Limitation: Requires active compliance

Third Eye (Our Model)

Core Innovation: Passive Gallery Parsing

Automated local face grouping, landmark alignment, and statistical Z-score anomaly tracking.

Advantage: Zero effort, 100% private, personalized baseline

RESEARCH

- **WHO:** Digital health can improve accessibility and people-centred healthcare
- **WHO:** AI in healthcare requires safety, privacy, equity and human oversight.
- **Computer Vision:** Face/landmark detection enables measurable visual-feature analysis.
- **Our Innovation:** Applies these technologies to personal-baseline change detection, not disease diagnosis..

REFERENCE

- WHO — Global Strategy on Digital Health 2020–2027
- WHO — Ethics and Governance of AI for Health
- WHO — Harnessing Artificial Intelligence for Health
- Google MediaPipe — Computer Vision & ML Solutions

THANK
YOU